

Project Title: E-Commerce Customer Churn Analysis

Problem Statement: In the realm of e-commerce, businesses face the challenge of understanding customer churn patterns to ensure customer satisfaction and sustained profitability. This project aims to delve into the dynamics of customer churn within an e-commerce domain, utilizing historical transactional data to uncover underlying patterns and drivers of churn. By analyzing customer attributes such as tenure, preferred payment modes, satisfaction scores, and purchase behavior, the project seeks to investigate and understand the dynamics of customer attrition and their propensity to churn. The ultimate objective is to equip e-commerce enterprises with actionable insights to implement targeted retention strategies and mitigate churn, thereby fostering long-term customer relationships and ensuring business viability in a competitive landscape.

Dataset Download: https://drive.google.com/uc?export=download&id=1iKKCze_Fpk2n_g3BIZBiSjcDFdFcEn3D

Project Steps and Objectives:

Data Cleaning: Handling Missing Values and Outliers

- loaded the dataset and displayed first five rows

The screenshot displays a database management interface with two main panels. The left panel, titled 'SCHEMAS', shows a tree view of the database structure. It includes a search bar labeled 'Filter objects' and a list of objects: 'ecomm', 'employee', and 'employees'. The right panel shows a list of SQL queries being executed. The queries are:

- 1 • DROP DATABASE IF EXISTS ecomm;
- 2 • CREATE DATABASE ecomm;
- 3 • USE ecomm;

Below these queries, a table schema for 'customer_churn' is shown. The table has the following columns and data types:

Column Name	Data Type
CustomerID	INT
Churn	BIT
Tenure	INT
PreferredLoginDevice	VARCHAR(10)
CityTier	INT
WarehouseToHome	INT
PreferredPaymentMode	INT
Gender	INT
HourSpendOnApp	INT
NumberOfDeviceRegistered	INT

The bottom panel shows the 'Output' of the SQL queries, which is empty.

5657 (55629,0,23,'Computer',3,9,'Credit Card','Male',4,5,'Laptop & Accessory',4,'Married',4,0,15,
 5658 (55630,0,8,'Mobile Phone',1,15,'Credit Card','Male',3,2,'Laptop & Accessory',3,'Married',4,4,6
 5659
 5660 • `select * from customer_churn limit 5;`
 5661

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HouseholdSize
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3
50002	1	NULL	Phone	1	8	UPI	Male	3
50003	1	NULL	Phone	1	30	Debit Card	Male	2
50004	1	0	Phone	3	15	Debit Card	Male	2

➤ **Impute mean for the following columns, and round off to the nearest integer if required:**
WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromLastYear, DaySinceLastOrder.

5680 • `UPDATE customer_churn`
 5681 `SET DaySinceLastOrder = ROUND((SELECT AVG(DaySinceLastOrder) FROM ecustomer_churn), 0)`
 5682 `WHERE DaySinceLastOrder IS NULL;`
 5683
 5684 • `SELECT * FROM customer_churn;`

Result Grid

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HouseholdSize
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3
50002	1	NULL	Phone	1	8	UPI	Male	3
50003	1	NULL	Phone	1	30	Debit Card	Male	3
50004	1	0	Phone	3	15	Debit Card	Male	3

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 137	02:54:58	USE ecomm	0 row(s) affected	0.000 sec
✓ 138	02:56:16	CREATE TABLE customer_churn(CustomerID ...	0 row(s) affected	0.032 sec
✓ 139	02:57:32		5630 row(s) affected Records: 5630 Duplicates: ...	0.219 sec
✗ 140	02:58:53	select * from customer_churn limit=5	Error Code: 1064. You have an error in your SQL s...	0.000 sec
✓ 141	02:59:01	select * from customer_churn limit 5	5 row(s) returned	0.016 sec / 0.000 sec
✓ 142	03:15:22	UPDATE customer_churn SET WarehouseToHo...	251 row(s) affected Rows matched: 251 Change...	0.031 sec
✗ 143	03:16:53	UPDATE customer_churn SET HourSpendOnAp...	Error Code: 1093. You can't specify target table 'c...	0.000 sec
✓ 144	03:21:05	SELECT ROUND(AVG(HourSpendOnApp)) INTO...	1 row(s) affected	0.000 sec
✓ 145	03:21:08	UPDATE customer_churn SET HourSpendOnAp...	2688 row(s) affected Rows matched: 5375 Chan...	0.062 sec
✓ 146	10:40:56	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
✓ 147	10:41:52	select * from customer_churn limit 5	5 row(s) returned	0.000 sec / 0.000 sec
✓ 148	10:42:12	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec

➤ **Impute mode for the following columns: Tenure, CouponUsed, OrderCount.**

5698 WHERE OrderCount IS NULL;

5699

5700 • SELECT * FROM customer_churn;

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	Hot
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3
50002	1	1	Phone	1	8	UPI	Male	3
50003	1	1	Phone	1	30	Debit Card	Male	3
50004	1	0	Phone	3	15	Debit Card	Male	3

customer_churn 5 x Apply Revert

Output :

#	Time	Action	Message	Duration / Fetch
✓ 142	03:15:22	UPDATE customer_churn SET WarehouseToHome...	251 row(s) affected Rows matched: 251 Change...	0.031 sec
✗ 143	03:16:53	UPDATE customer_churn SET HourSpendOnAp...	Error Code: 1093. You can't specify target table 'c...	0.000 sec
✓ 144	03:21:05	SELECT ROUND(AVG(HourSpendOnApp)) INTO...	1 row(s) affected	0.000 sec
✓ 145	03:21:08	UPDATE customer_churn SET HourSpendOnAp...	2688 row(s) affected Rows matched: 5375 Chan...	0.062 sec
✓ 146	10:40:56	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
✓ 147	10:41:52	select * from customer_churn limit 5	5 row(s) returned	0.000 sec / 0.000 sec
✓ 148	10:42:12	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
✓ 149	10:56:15	UPDATE customer_churn SET Tenure = (SELEC...	264 row(s) affected Rows matched: 264 Change...	0.141 sec
✓ 150	10:56:20	UPDATE customer_churn SET CouponUsed = (...	256 row(s) affected Rows matched: 256 Change...	0.125 sec
✓ 151	10:56:23	UPDATE customer_churn SET OrderCount = (S...	258 row(s) affected Rows matched: 258 Change...	0.125 sec
✓ 152	10:56:46	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec

➤ Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

5700 • SELECT * FROM customer_churn;

5701

5702 -- Handle outliers in the 'WarehouseToHome' column by deleting rows where the

5703

5704 • DELETE FROM customer_churn WHERE WarehouseToHome > 100;

5705

Output :

#	Time	Action	Message
✗ 143	03:16:53	UPDATE customer_churn SET HourSpendOnAp...	Error Code: 1093. You can't specify target table 'c...
✓ 144	03:21:05	SELECT ROUND(AVG(HourSpendOnApp)) INTO...	1 row(s) affected
✓ 145	03:21:08	UPDATE customer_churn SET HourSpendOnAp...	2688 row(s) affected Rows matched: 5375 Chan...
✓ 146	10:40:56	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
✓ 147	10:41:52	select * from customer_churn limit 5	5 row(s) returned
✓ 148	10:42:12	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
✓ 149	10:56:15	UPDATE customer_churn SET Tenure = (SELEC...	264 row(s) affected Rows matched: 264 Change...
✓ 150	10:56:20	UPDATE customer_churn SET CouponUsed = (...	256 row(s) affected Rows matched: 256 Change...
✓ 151	10:56:23	UPDATE customer_churn SET OrderCount = (S...	258 row(s) affected Rows matched: 258 Change...
✓ 152	10:56:46	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
✓ 153	11:01:00	DELETE FROM customer_churn WHERE Wareh...	2 row(s) affected

Data Cleaning: Dealing with Inconsistencies:

➤ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferredOrderCat' column with “Mobile Phone” to ensure uniformity.

```

5711 • UPDATE customer_churn SET PreferredOrderCat = 'Mobile Phone'
5712 WHERE PreferredOrderCat = 'Mobile';
5713
5714 • SELECT PreferredLoginDevice,PreferredOrderCat FROM customer_churn;

```

PreferredLoginDevice	PreferredOrderCat
Phone	Mobile
Computer	Mobile
Mobile Phone	Laptop & Accessory
Computer	Laptop & Accessory
Mobile Phone	Mobile Phone

customer_churn 6 x

Output

#	Time	Action	Message	Du
✓ 144	03:21:05	SELECT ROUND(AVG(HourSpendOnApp)) INTO...	1 row(s) affected	0.0
✓ 145	03:21:08	UPDATE customer_churn SET HourSpendOnAp...	2688 row(s) affected Rows matched: 5375 Chan...	0.0
✓ 146	10:40:56	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.0
✓ 147	10:41:52	select * from customer_churn limit 5	5 row(s) returned	0.0
✓ 148	10:42:12	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.0
✓ 149	10:56:15	UPDATE customer_churn SET Tenure = (SELEC...	264 row(s) affected Rows matched: 264 Change...	0.1
✓ 150	10:56:20	UPDATE customer_churn SET CouponUsed = (...	256 row(s) affected Rows matched: 256 Change...	0.1
✓ 151	10:56:23	UPDATE customer_churn SET OrderCount = (S...	258 row(s) affected Rows matched: 258 Change...	0.1
✓ 152	10:56:46	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.0
✓ 153	11:01:00	DELETE FROM customer_churn WHERE Wareh...	2 row(s) affected	0.0
✓ 154	11:05:39	SELECT PreferredLoginDevice,PreferredOrderCat ...	1000 row(s) returned	0.0

➤ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

5721 • UPDATE customer_churn SET PreferredPaymentMode = 'Credit Card'

5722 WHERE PreferredPaymentMode = 'CC';

5723

5724 • SELECT PreferredPaymentMode FROM customer_churn;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

PreferredPaymentMode
Credit Card
Debit Card
Cash on Delivery
Credit Card
E wallet
Debit Card

customer_churn 8 x

Output

Action Output

#	Time	Action	Message
✓ 148	10:42:12	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
✓ 149	10:56:15	UPDATE customer_churn SET Tenure = (SELEC...	264 row(s) affected Rows matched: 264 Change...
✓ 150	10:56:20	UPDATE customer_churn SET CouponUsed = (...	256 row(s) affected Rows matched: 256 Change...
✓ 151	10:56:23	UPDATE customer_churn SET OrderCount = (S...	258 row(s) affected Rows matched: 258 Change...
✓ 152	10:56:46	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned
✓ 153	11:01:00	DELETE FROM customer_churn WHERE Wareh...	2 row(s) affected
✓ 154	11:05:39	SELECT PreferredLoginDevice,PreferredOrderCat ...	1000 row(s) returned
✓ 155	11:10:18	SELECT PreferredPaymentMode FROM customer...	1000 row(s) returned
✓ 156	11:10:52	UPDATE customer_churn SET PreferredPayment...	365 row(s) affected Rows matched: 365 Change...
✓ 157	11:10:54	UPDATE customer_churn SET PreferredPayment...	273 row(s) affected Rows matched: 273 Change...
✓ 158	11:10:57	SELECT PreferredPaymentMode FROM customer...	1000 row(s) returned

Data Transformation:Data Transformation:

➤ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

➤ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

```

5730 • ALTER TABLE customer_churn RENAME COLUMN PreferredOrderCat TO PreferredOrderCat;
5731 • ALTER TABLE customer_churn RENAME COLUMN HourSpendOnApp TO HoursSpentOnApp;
5732
5733 • SELECT PreferredOrderCat,HoursSpentOnApp FROM customer_churn;

```

PreferredOrderCat	HoursSpentOnApp
Laptop & Accessory	3
Mobile	3
Mobile	3
Laptop & Accessory	3
Mobile	NULL
Mobile Phone	3

customer_churn 9 x

Output

#	Time	Action	Message	Duration
✓ 153	11:01:00	DELETE FROM customer_churn WHERE Wareh...	2 row(s) affected	0.04
✓ 154	11:05:39	SELECT PreferredLoginDevice,PreferredOrderCat ...	1000 row(s) returned	0.00
✓ 155	11:10:18	SELECT PreferredPaymentMode FROM customer...	1000 row(s) returned	0.00
✓ 156	11:10:52	UPDATE customer_churn SET PreferredPayment...	365 row(s) affected Rows matched: 365 Change...	0.14
✓ 157	11:10:54	UPDATE customer_churn SET PreferredPayment...	273 row(s) affected Rows matched: 273 Change...	0.14
✓ 158	11:10:57	SELECT PreferredPaymentMode FROM customer...	1000 row(s) returned	0.01
✓ 159	11:16:46	ALTER TABLE customer_churn RENAME COLU...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.07
✗ 160	11:16:49	ALTER TABLE customer_churn RENAME COLU...	Error Code: 1064. You have an error in your SQL s...	0.01
✗ 161	11:16:53	SELECT PreferredOrderCat,HoursSpentOnApp F...	Error Code: 1054. Unknown column 'HoursSpent...	0.01
✓ 162	11:17:12	ALTER TABLE customer_churn RENAME COLU...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.07
✓ 163	11:17:15	SELECT PreferredOrderCat,HoursSpentOnApp F...	1000 row(s) returned	0.00

Data Transformation:Creating New Columns:

➤ Create a new column named ‘ComplaintReceived’ with values "Yes" if the corresponding value in the ‘Complain’ is 1, and "No" otherwise.

➤ Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

5743 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(10);

5744 • UPDATE customer_churn SET ChurnStatus = CASE WHEN Churn = 1 THEN 'Churned' ELSE 'Active' END;

5745

5746 • SELECT ComplaintReceived,ChurnStatus FROM customer_churn;

Result Grid

ComplaintReceived	ChurnStatus
No	Churned
Yes	Churned
Yes	Churned
No	Churned
No	Churned

customer_churn 10 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 159	11:16:46	ALTER TABLE customer_churn RENAME COLU...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.078 sec
✗ 160	11:16:49	ALTER TABLE customer_churn RENAME COLU...	Error Code: 1064. You have an error in your SQL s...	0.016 sec
✗ 161	11:16:53	SELECT PreferredOrderCat,HoursSpentOnApp F...	Error Code: 1054. Unknown column 'HoursSpent...	0.015 sec
✓ 162	11:17:12	ALTER TABLE customer_churn RENAME COLU...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.078 sec
✓ 163	11:17:15	SELECT PreferredOrderCat,HoursSpentOnApp F...	1000 row(s) returned	0.000 sec / 0.000 sec
✗ 164	11:26:32	SELECT ComplaintReceived,ChurnStatus FROM ...	Error Code: 1054. Unknown column 'ComplaintRe...	0.000 sec
✓ 165	11:26:37	ALTER TABLE customer_churn ADD COLUMN C...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.078 sec
✓ 166	11:26:45	UPDATE customer_churn SET ComplaintReceive...	5628 row(s) affected Rows matched: 5628 Chan...	1.125 sec
✓ 167	11:26:52	ALTER TABLE customer_churn ADD COLUMN C...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.078 sec
✓ 168	11:26:55	UPDATE customer_churn SET ChurnStatus = CA...	5628 row(s) affected Rows matched: 5628 Chan...	1.063 sec
✓ 169	11:26:58	SELECT ComplaintReceived,ChurnStatus FROM ...	1000 row(s) returned	0.000 sec / 0.000 sec

Data Transformation:Column Dropping:

➤ Drop the columns "Churn" and "Complain" from the table.

PreferredPaymentMod
Gender
HoursSpentOnApp
NumberOfDeviceRegis
PreferredOrderCat
SatisfactionScore
MaritalStatus
NumberOfAddress
OrderAmountHikeFor
CouponUsed
OrderCount
DaySinceLastOrder
CashbackAmount
ComplaintReceived

5742

5743 • ALTER TABLE customer_churn ADD COLUMN ChurnStatus VARCHAR(10);

5744 • UPDATE customer_churn SET ChurnStatus = CASE WHEN Churn = 1 THEN 'Churned' ELSE 'A

5745

5746 • SELECT ComplaintReceived,ChurnStatus FROM customer_churn;

5747

5748 -- -- Data Transformation: Drop the Columns Churn and Complain from the table --

5749

5750 • ALTER TABLE customer_churn DROP COLUMN Churn;

5751 • ALTER TABLE customer_churn DROP COLUMN Complain;

5752

5753 • SELECT Churn,Complain FROM customer_churn;

chema: schooldb

Output

Action Output

#	Time	Action	Message	Dura
✓ 163	11:17:15	SELECT PreferredOrderCat,HoursSpentOnApp F...	1000 row(s) returned	0.001
✗ 164	11:26:32	SELECT ComplaintReceived,ChurnStatus FROM ...	Error Code: 1054. Unknown column 'ComplaintRe...	0.001
✓ 165	11:26:37	ALTER TABLE customer_churn ADD COLUMN C...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.071
✓ 166	11:26:45	UPDATE customer_churn SET ComplaintReceive...	5628 row(s) affected Rows matched: 5628 Chan...	1.121
✓ 167	11:26:52	ALTER TABLE customer_churn ADD COLUMN C...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.071
✓ 168	11:26:55	UPDATE customer_churn SET ChurnStatus = CA...	5628 row(s) affected Rows matched: 5628 Chan...	1.061
✓ 169	11:26:58	SELECT ComplaintReceived,ChurnStatus FROM ...	1000 row(s) returned	0.001
✗ 170	11:30:34	ALTER TABLE ecomm DROP COLUMN Churn	Error Code: 1146. Table 'ecomm.ecomm' doesn't e...	0.001
✓ 171	11:31:00	ALTER TABLE customer_churn DROP COLUMN...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.061
✓ 172	11:31:03	ALTER TABLE customer_churn DROP COLUMN...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.061
✗ 173	11:31:56	SELECT Churn,Complain FROM customer_churn ...	Error Code: 1054. Unknown column 'Churn' in fiel...	0.001

ct Info Session

Data Exploration and Analysis

➤ Retrieve the count of churned and active customers from the dataset.

```
5756
5757 • SELECT ChurnStatus, COUNT(*) AS CustomerCount FROM customer_churn GROUP BY ChurnStatus;
```

ChurnStatus	CustomerCount
Churned	948
Active	4680

Result 11 x Read Only

➤ Display the average tenure and total cashback amount of customers who churned.

```
5761 • SELECT AVG(Tenure) AS Avg_Tenure, SUM(CashbackAmount) AS Total_Cashback
5762 FROM customer_churn WHERE ChurnStatus = 'Churned';
```

Avg_Tenure	Total_Cashback
3.1762	152030

➤ Determine the percentage of churned customers who complained.

```
5766 • SELECT
5767 (SUM(CASE WHEN ComplaintReceived = 'Yes' THEN 1 ELSE 0 END) * 100.0) / COUNT(*) AS Complaint_P
5768 FROM customer_churn WHERE ChurnStatus = 'Churned';
```

Complaint_Percentage
53.58650

Result Grid Form Editor

➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

```
5772 • SELECT CityTier, COUNT(*) AS ChurnCount FROM customer_churn
5773 WHERE ChurnStatus = 'Churned' AND PreferredOrderCat = 'Laptop & Accessory'
5774 GROUP BY CityTier ORDER BY ChurnCount DESC LIMIT 1;
5775
```

CityTier	ChurnCount
3	150

Result Grid Fetch rows

➤ Identify the most preferred payment mode among active customers.

```
5777
5778 • SELECT PreferredPaymentMode, COUNT(*) AS ActiveCount FROM customer_churn
5779 WHERE ChurnStatus = 'Active' GROUP BY PreferredPaymentMode
5780 ORDER BY ActiveCount DESC LIMIT 1;
5781
```

PreferredPaymentMode	ActiveCount
Debit Card	1956

➤ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

```
5783
5784 • SELECT SUM(OrderAmountHikeFromlastYear) AS TotalOrderHike FROM customer_churn
5785 WHERE MaritalStatus = 'Single' AND PreferredOrderCat = 'Mobile Phone';
5786
```

TotalOrderHike
7738

➤ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

```
5789 • SELECT AVG(NumberOfDeviceRegistered) AS Avg_Devices FROM customer_churn
5790 WHERE PreferredPaymentMode = 'UPI';
5791
```

Avg_Devices
3.7174

➤ Determine the city tier with the highest number of customers.

```
5794 • SELECT CityTier, COUNT(*) AS TotalCustomers FROM customer_churn GROUP BY CityTier
5795 ORDER BY TotalCustomers DESC LIMIT 1;
```

CityTier	TotalCustomers
1	3666

➤ Identify the gender that utilized the highest number of coupons.

5798

5799 • `SELECT Gender, SUM(CouponUsed) AS TotalCoupons_Used FROM customer_churn`

5800 `GROUP BY Gender ORDER BY TotalCoupons_Used DESC LIMIT 1;`

5801

Result Grid

Gender	TotalCoupons_Used
Male	5629

➤ List the number of customers and the maximum hours spent on the app in each preferred order category.

5802

5803

5804 • `SELECT PreferredOrderCat, COUNT(*) AS CustomerCount, MAX(HoursSpentOnApp) AS MaxHoursSpent`

5805 `FROM customer_churn GROUP BY PreferredOrderCat;`

5806

Result Grid

PreferredOrderCat	CustomerCount	MaxHoursSpent
Laptop & Accessory	2050	3
Mobile	808	3
Mobile Phone	1270	3
Others	264	3
Fashion	826	3

➤ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

5808

5809 • `SELECT SUM(OrderCount) AS TotalOrders FROM customer_churn`

5810 `WHERE PreferredPaymentMode = 'Credit Card' AND`

5811 `SatisfactionScore = (SELECT MAX(SatisfactionScore) FROM customer_churn);`

Result Grid

TotalOrders
1122

➤ What is the average satisfaction score of customers who have complained?

5814

5815 • `SELECT AVG(SatisfactionScore) AS AvgSatisfaction_Score`

5816 `FROM customer_churn WHERE ComplaintReceived = 'Yes';`

Result Grid

AvgSatisfaction_Score
2.9988

➤ List the preferred order category among customers who used more than 5 coupons.

5819

5820 • `SELECT DISTINCT PreferredOrderCat FROM customer_churn WHERE CouponUsed > 5;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

PreferredOrderCat
Others
Laptop & Accessory
Fashion
Mobile
Grocery

➤ List the top 3 preferred order categories with the highest average cashback amount.

5824 • `SELECT PreferredOrderCat, AVG(CashbackAmount) AS Avg_Cashback`
 5825 `FROM customer_churn`
 5826 `GROUP BY PreferredOrderCat ORDER BY Avg_Cashback DESC LIMIT 3;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#) | Fetch rows:

PreferredOrderCat	Avg_Cashback
Others	304.4545
Grocery	266.2366
Fashion	210.4031

➤ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

5829

5830 • `SELECT PreferredPaymentMode FROM customer_churn`
 5831 `GROUP BY PreferredPaymentMode HAVING AVG(Tenure) = 10 AND SUM(OrderCount) > 500;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

PreferredPaymentMode

➤ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category

5841 `END AS DistanceCategory, ChurnStatus, COUNT(*) AS Customer_Count`
 5842 `FROM customer_churn GROUP BY DistanceCategory, ChurnStatus;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

DistanceCategory	ChurnStatus	Customer_Count
Close Distance	Churned	265
Far Distance	Churned	498
Moderate Distance	Churned	184
Close Distance	Active	1696
Moderate Distance	Active	1106
Far Distance	Active	1071

Result 27 x

➤ List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

5845
5846 • `SELECT * FROM customer_churn`
5847 `WHERE MaritalStatus = 'Married' AND CityTier = 1`
5848 `AND OrderCount > (SELECT AVG(OrderCount) FROM customer_churn);`

Result Grid

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpent
50049	3	Computer	1	15	Credit Card	Male	10
50119	13	Phone	1	30	Cash on Delivery	Male	3
50132	15	Phone	1	16	Credit Card	Female	3
50133	13	Mobile Phone	1	8	Credit Card	Male	3
50157	2	Mobile Phone	1	28	Debit Card	Female	2

customer_churn 28 x

➤ a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990

- Created the table

5851
5852 • `CREATE TABLE customer_returns (`
5853 `ReturnID INT,`
5854 `CustomerID INT,`
5855 `ReturnDate DATE,`
5856 `RefundAmount INT`
5857 `);`
5858

ecomm
Tables
customer_churn
customer_returns
Columns
ReturnID
CustomerID
ReturnDate
RefundAmount
Indexes

- Inserted values

5868

5869 • `SELECT * FROM customer_returns;`

5870

Result Grid | Filter Rows: | Export:

	ReturnID	CustomerID	ReturnDate	RefundAmount
▶	1001	50022	2023-01-01	2130
	1002	50316	2023-01-23	2000
	1003	51099	2023-02-14	2290
	1004	52321	2023-03-08	2510
	1005	52928	2023-03-20	1006
	1006	53749	2023-04-17	3000

customer_returns 29 x

b) Display the return details along with the customer details of those who have churned and have made complaints.

5873 • `SELECT * FROM customer_returns JOIN customer_churn`

5874 `ON customer_returns.CustomerID=customer_churn.CustomerID`

5875 `WHERE ChurnStatus = 'Churned' and ComplaintReceived = 'Yes';`

5876

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	ReturnID	CustomerID	ReturnDate	RefundAmount	CustomerID	Tenure	PreferredLoginDevice	CityTier	Wareho.
▶	1002	50316	2023-01-23	2000	50316	0	Computer	2	29
	1004	52321	2023-03-08	2510	52321	18	Mobile Phone	3	19
	1006	53749	2023-04-17	3000	53749	1	Mobile Phone	3	31